

LESSON 1.6

WRITING AND SOLVING TWO-STEP EQUATIONS



LESSON INTRODUCTION

MA.7.AR.2.2 Write and solve two-step equations in one variable within a mathematical or real-world context, where all terms are rational numbers.

LEARNING TARGETS

- I can solve a two-step equation within a mathematical or real-world context.
- I can write a two-step equation given a real-world context.

BENCHMARK COHERENCE

BUILDING ON

MA.6.AR.2.3 Write and solve one-step equations in one variable within a mathematical or real-world context using multiplication and division, where all terms and solutions are integers.

WORKING TOWARD

MA.8.AR.2.2 Solve two-step linear inequalities in one variable and represent solutions algebraically and graphically.

ESSENTIAL QUESTIONS

- Which properties of equality did you use to solve the equation?
- How can you check your solution?
- What two methods can you use to solve an equation of the form $a(b+c) = d$? Which method do you prefer?
- What steps do you take to write an equation from a real-world context?

LESSON NARRATIVE

Student initially review the properties of equality. Then, they apply them to solve two-step equations and use substitution to check their answers. Finally, students write and solve two-step equations given real-world contexts.

COMPONENT SUMMARY

1.6.1 WARM-UP (~5 MIN.)

Estimate time needed to create paper clip chain

1.6.3 TRY IT (5-10 MIN.)

Solve two-step equations and check solutions

1.6.5 GUIDED INSTRUCTION (10 MIN.)

Write and solve equation given real-world context

1.6.7 WRAP-UP (~5 MIN.)

Identify and correct errors in samples of student work

1.6.2 GUIDED PRACTICE (10 MIN.)

Write numeric examples and apply properties of equality

1.6.4 COLLABORATION (10 MIN.)

Explore two methods for applying distributive property

1.6.6 TRY IT (10 MIN.)

Solve equations and match real-world context to an equation

RESOURCES

GLOSSARY

Key Terms:

- N/A

Additional Terms:

- Symmetric Property of Equality
- Addition Property of Equality
- Subtraction Property of Equality
- Multiplication Property of Equality
- Division Property of Equality
- Distributive Property

MATERIALS

- (Optional) Highlighters
- (Optional) Paper Clips
- (Optional) Hands-On Equations
- (Optional) Algebra Tiles

ADDITIONAL PREPARATION

- Consider showing students how to concretely model the equations using a Hands-On Equations kit or Algebra Tiles during 1.6.2 Guided Practice. If you do not have the concrete manipulatives, consider using virtual manipulatives.

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may not understand that they must apply inverse operations to both sides of the equation.
- Students may not understand that when applying the distributive property, the number outside of the parentheses must be multiplied by each term inside of the parentheses.
- Students may not understand that the inverse operation of multiplying by a fraction is multiplying by its reciprocal, which will be a whole number if the fraction was a unit fraction.

THINGS TO CONSIDER

- Consider creating and displaying the properties on an Anchor Chart or Word Wall for students to reference.
- Ask students what they can multiply the fraction $\frac{a}{b}$ by to obtain a product of one. This will help students understand why they should multiply both sides of the equation by the reciprocal of the fraction.
- If an equation contains fractions and decimals, prompt students to either convert the fractions to decimals or vice versa.

LITERACY AND LANGUAGE ROUTINES

Think-Pair-Share

In 1.6.4 Collaboration, have students initially solve the equation using method 1 and method 2 independently. Next, prompt students to discuss the differences between the methods with a partner. Finally, ask a variety of students to share which method they prefer and why.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.7.1 Real-World Contexts

In 1.6.5 Guided Instruction, students write and solve two-step equations based on real-world contexts. In 1.6.6 Try It, students match a two-step equation to the real-world context that it represents. These components ignite student interest and help students gain an appreciation for the utility of the mathematics they are learning.

DIFFERENTIATE INSTRUCTION

In 1.6.5 Guided Instruction and 1.6.6 Try It, provide a visual entry-point by showing students how to draw a diagram of the situation. Partner discussions during 1.6.4 Collaboration and 1.6.5 Guided Instruction provide students with an auditory entry-point. Provide a kinesthetic entry-point during 1.6.1 Warm-Up by providing students with paper clips, and during 1.6.2 Guided Practice by showing students how to concretely model the equations using a Hands-On Equations kit or Algebra Tiles.

Additional differentiation opportunities are denoted throughout the lesson guide.

Cost per
ticket=



$$\begin{array}{ccccc} \text{Ticket Icon} & \text{Ticket Icon} & \text{Ticket Icon} & \text{Ticket Icon} & \text{Ticket Icon} \\ -13.25 & -13.25 & -13.25 & -13.25 & -13.25 \end{array} = 78.70$$

1.6.1 WARM-UP: ESTIMATION

DIFFERENTIATE INSTRUCTION

Consider providing students with paper clips to provide visual and kinesthetic entry-points. Encourage students to time how long it takes to create a chain of ten clips. Then, prompt them to use this value to estimate how long it would take to create a chain of one hundred clips.



1.6.1 Warm-Up: Estimation

The picture shows 100 #4 size paper clips with four clips chained together.



- Complete the statements by estimating how long it would take you to chain 100 paper clips together.
 - Too Low: 30 seconds
 - Too High: 6 hours
 - Just Right: 20 minutes

Answers will vary.



1.6.2 Guided Practice: Properties of Equality

Properties of equality can be used to solve equations.
1. Complete the table with a numeric example of each property of equality.

Property	Algebraic Statement	Numeric Example
Symmetric Property of Equality	If $a = b$, then $b = a$.	$7 = x$, then $x = 7$
Addition Property of Equality	If $a = b$, then $a + c = b + c$.	If $x = 5$, then $x + 3 = 5 + 3$
Subtraction Property of Equality	If $a = b$, then $a - c = b - c$.	If $x = 9$, then $x - 4 = 9 - 4$
Multiplication Property of Equality	If $a = b$, then $a \cdot c = b \cdot c$.	If $x = 2$, then $x \cdot 7 = 2 \cdot 7$
Division Property of Equality	If $a = b$ and $c \neq 0$, then $a \div c = b \div c$.	If $x = 12$, then $x \div 4 = 12 \div 4$

2. Which property justifies the equation $4m + 7.1 = 24.38$ is equivalent to $24.38 = 4m + 7.1$?
Symmetric Property of Equality
3. Solve $-8.1 = \frac{z}{-3} - \frac{2}{5}$ by applying properties of equality.
- a. Fill in each box with an operation and each blank with a value.

$$\begin{aligned} -8.1 &= \frac{z}{-3} - \frac{2}{5} \\ -8.1 \boxed{+} \frac{2}{5} &= \frac{z}{-3} - \frac{2}{5} \boxed{+} \frac{2}{5} \\ \frac{-7.7}{-3} &= \frac{z}{-3} \\ \frac{-3}{-3} \boxed{\times} \frac{-7.7}{-3} &= \frac{-3}{-3} \boxed{\times} \frac{z}{-3} \\ \frac{23.1}{-3} &= z \end{aligned}$$

1.6.2 GUIDED PRACTICE: PROPERTIES OF EQUALITY

TEACHER MOVES

Collect and Display
Listen for and collect a variety of students' numeric examples. Use them to create an Anchor Chart. Display the chart for students to reference in subsequent components.

ENRICHMENT

Which property of equality did you use at each step?

1.6.2 GUIDED PRACTICE: PROPERTIES OF EQUALITY (CONTINUED)

DIFFERENTIATE INSTRUCTION

Provide kinesthetic and visual entry-points by showing students how to concretely model the equations using a Hands-On Equations kit or Algebra Tiles. If you do not have the concrete manipulatives, consider using virtual manipulatives.

- b. Use substitution in the original equation to verify the solution.

$$\begin{aligned} -8.1 &= \frac{x}{-3} - \frac{2}{5} \\ -8.1 &= \frac{23.1}{-3} - \frac{2}{5} \\ -8.1 &= -7.7 - 0.4 \\ -8.1 &= -8.1 \end{aligned}$$

4. Solve $5x + 4.25 = -3\frac{1}{4}$ by applying properties of equality.

- a. Fill in each box with an operation and each blank with a value for the first step in solving the equation. Then, finish solving the equation on your own.

$$\begin{aligned} 5x + 4.25 &= -3\frac{1}{4} \\ 5x + 4.25 \boxed{-} \underline{4.25} &= -3\frac{1}{4} \boxed{-} \underline{4.25} \\ 5x &= \underline{-7.5} \\ x &= \underline{-1.5} \end{aligned}$$

- b. Use substitution in the original equation to verify the solution.

$$\begin{aligned} 5x + 4.25 &= -3\frac{1}{4} \\ 5(-1.5) + 4.25 &= -3.25 \\ -7.5 + 4.25 &= -3.25 \\ -3.25 &= -3.25 \end{aligned}$$



1.6.3 Try It: Solving Two-Step Equations

- Solve each equation by applying properties of equality. Use substitution to verify the solutions for each equation.

$\begin{array}{r} -4w - 7 = -11.2 \\ +7 \quad +7 \\ \hline -4w = -4.2 \\ -4 \quad -4 \\ \hline w = 1.05 \end{array}$	$\begin{array}{r} \frac{g}{8} + \frac{1}{2} = 0 \\ -\frac{1}{2} \quad -\frac{1}{2} \\ \hline 8 \times \frac{g}{8} = -\frac{1}{2} \times 8 \\ g = -4 \end{array}$
$\begin{array}{r} -4(1.05) - 7 = -11.2 \\ -4.2 - 7 = -11.2 \\ -11.2 = -11.2 \end{array}$	$\begin{array}{r} -\frac{4}{8} + \frac{1}{2} = 0 \\ -0.5 + 0.5 = 0 \\ 0 = 0 \end{array}$
$\begin{array}{r} 10.5 = \frac{r}{6} - \frac{3}{4} \\ +0.75 \quad +0.75 \\ \hline 6 \times 11.25 = \frac{r}{6} \times 6 \\ r = 67.5 \end{array}$	$\begin{array}{r} -\frac{12}{5} = 5m + 8.2 \\ -8.2 \quad -8.2 \\ \hline -10.6 = 5m \\ \frac{-10.6}{5} = \frac{5m}{5} \\ m = -2.12 \end{array}$
$\begin{array}{r} 10.5 = \frac{67.5}{6} - \frac{3}{4} \\ 10.5 = 11.25 - 0.75 \\ 10.5 = 10.5 \end{array}$	$\begin{array}{r} -\frac{12}{5} = 5(-2.12) + 8.2 \\ -2.4 = -10.6 + 8.2 \\ -2.4 = -2.4 \end{array}$

1.6.3 TRY IT: SOLVING TWO-STEP EQUATIONS

TEACHER MOVES

Take Turns

Consider allowing students to work with a partner and either take turns performing the steps in each problem or alternate explaining every other problem. Prompt students to reference the properties of equality Anchor Chart and resolve any disagreements through discussion.

ENRICHMENT

What can you multiply the fraction a/b by to obtain a product of one?

1.6.4 COLLABORATION: USING THE DISTRIBUTIVE PROPERTY

TEACHER MOVES

Think-Pair-Share

Have students initially solve the equation using method 1 and method 2 independently. Next, prompt students to discuss the differences between the methods with a partner. Finally, ask a variety of students to share which method they prefer and why.

DIFFERENTIATE INSTRUCTION

Partner discussions during this component provide students with an auditory entry-point.



1.6.4 Collaboration: Using the Distributive Property

Equations can be solved in more than one way. When an equation is written in the form $a(b + c) = d$, it can be solved by either distributing a first or by dividing both sides of the equation by a first.

1. The equation $63 = 3(6 - p)$ can be solved using either method.
 - a. Solve the equation $63 = 3(6 - p)$ using each method.

Method 1 Apply the distributive property first.	Method 2 Divide both sides by the factor being distributed first.
$ \begin{array}{r} 63 = 3(6 - p) \\ 63 = 18 - 3p \\ -18 \quad -18 \\ \hline 45 = -3p \\ -3 \quad -3 \\ \hline p = -15 \end{array} $	$ \begin{array}{r} 63 = 3(6 - p) \\ \frac{3}{3} \quad \frac{3}{3} \\ 21 = 6 - p \\ -6 \quad -6 \\ \hline 15 = -p \\ -1 \quad -1 \\ \hline p = -15 \end{array} $

- b. Verify that the solution is correct using substitution.

$$\begin{array}{l}
 63 = 3(6 - (-15)) \\
 63 = 3(6 + 15) \\
 63 = 3(21) \\
 63 = 63
 \end{array}$$



1.6.5 Guided Instruction: Writing a Two-Step Equation from a Real-World Context

When writing equations to represent a real-world context, it is helpful to start by considering the known and unknown quantities involved in the situation.

1. Darius is draining his pool for resurfacing. The pool began with 13,650 gallons of water and it is draining at a rate of 640 gallons per hour. How long has the pool been draining if currently there are 9,330 gallons in the pool?

- a. Complete the table to describe the known and unknown quantities in this situation.

Known Quantities	Unknown Quantities
<i>Start: 13,650 gallons -640 gallons per hour Current: 9,330 gallons</i>	<i>Length of time the pool draining</i>

- b. With your partner, discuss how the quantities are related in this situation. Summarize your discussion.

The number of gallons the pool began with, and the current amount can be used to determine the change in gallons. The pool drained 640 gallons per hour, so this coupled with the change in gallons can be used to find the length of time the pool has been draining.



When writing equations to represent real-world contexts, **any variables used must be defined** to identify which quantity each variable in the equation represents.

- c. Define the unknown quantity using a variable.
t, the length of time the pool is draining.
- d. Write an equation that could be used to find the unknown quantity using the variable you defined.
 $13,650 - 640t = 9,330$ or $9,330 + 640t = 13,650$
- e. Solve the equation.

$$\frac{640t}{640} = \frac{4320}{640}$$
 $t = 6.75$
- f. Explain what the solution means in this context.
The pool has been draining for 6.75 (6 hours and 45 minutes) hours.

1.6.5 GUIDED INSTRUCTION: WRITING A TWO-STEP EQUATION FROM A REAL-WORLD CONTEXT

TEACHER MOVES

Students write and solve two-step equations based on real-world contexts. The real-world applications will help ignite student interest. It will also help students gain an appreciation for the utility of the mathematics they are learning.

DIFFERENTIATE INSTRUCTION

Provide a visual entry-point by encouraging students to highlight or circle the important quantities in the word problem.

ENRICHMENT

Consider prompting students with the following:

- How can you check your solution?
- What units should you include with your final answer?
- Given the context, does your answer make sense?
- What would be an answer that would not make sense?

1.6.6 TRY IT: RELATING EQUATIONS TO THEIR CONTEXTS

DIFFERENTIATE INSTRUCTION

Provide a visual entry-point by showing students how to draw a diagram of the situation.

!!!

ENRICHMENT

What is the difference between Equations A and B?



1.6.6 Try It: Relating Equations to Their Contexts

Work with your partner to complete the following.

1. Two equations are shown.
 - a. Solve both equations. Show your work.

Equation A	Equation B
$78.70 = 5(y - 13.25)$ $78.70 = 5y - 66.25$ $+66.25 \quad +66.25$ $\underline{144.95 = 5y}$ $\quad 5 \quad 5$ $y = 28.99$	$5y - 13.25 = 78.70$ $\quad +13.25 \quad +13.25$ $\quad \underline{5y = 91.95}$ $\quad 5 \quad 5$ $y = 18.39$

- b. Five friends are going to a concert. They have a coupon for \$13.25 off of each ticket. They pay \$78.70 for all their tickets. Explain which equation represents this situation.
 $78.70 = 5(y - 13.25)$
Equation A represents this situation because the 5 on the outside parentheses represents the number of friends and the inside of the parentheses represents the ticket cost minus the discount.
 - c. Explain what the solution to the equation that correctly models this situation means.
Each ticket costs \$28.99 before the coupon.



1.6.7 Wrap-Up: Error Analysis

Two students solved an equation. One student made a mistake. Their work is shown below.

Caroline	Ivy
$\frac{1}{3}x + 7 = 40$	$\frac{1}{3}x + 7 = 40$
$3\left(\frac{1}{3}x + 7 = 40\right)$	$\frac{1}{3}x = 33$
$x + 21 = 120$	$x = 11$
$x = 99$	

- Which student made an error when solving the equation?
Ivy
- Explain what error the student made.
After subtracting 7 from both sides of equation, Ivy should have multiplied both sides of the equation by 3 instead of dividing by 3.

1.6.7 WRAP-UP: ERROR ANALYSIS

DIFFERENTIATE INSTRUCTION

Consider prompting students to first solve the equation themselves before they look at the sample students’ work. Then, students can compare their work to the sample students’ work. Also, consider providing students with sentence stems or answer choices to help them compose their response to question 2.